Banking Fraud Detection

Using Isolation Forest

Finance and Data Science — Final Project Report

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Outline

1. Introduction
2. Theoretical Background
3. The Isolation Forest Algorithm
4. Dataset and Methodology
5. R Implementation and Results
6. Limitations and Future Work
7. Conclusion

General Context

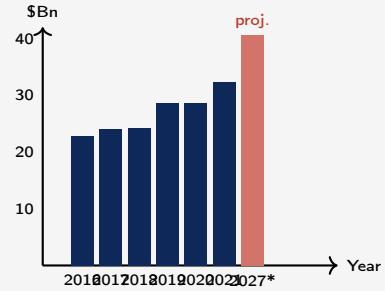
The Banking Fraud Challenge

- Rapid digital transformation of banking services
- Surge in online and mobile payments
- New avenues for criminal exploitation
- **Global losses: $32.34 Bn in 2021**
- Projected: >**$40 Bn by 2027**

Core Problem

Fraud represents **less than 0.2%** of all transactions — extreme class imbalance.

Global Card Fraud Losses ($Bn)



Problem Statement and Objectives

Research Question

How can *Machine Learning* techniques effectively detect fraudulent banking transactions in highly imbalanced datasets?

Objectives

1. Identify anomalous (fraudulent) transactions
2. Apply an unsupervised method suited to imbalance
3. Evaluate the effectiveness of Isolation Forest
4. Contribute to real-world financial security

Adopted Approach

- **Algorithm:** Isolation Forest (IF)
- **Language:** R (`isotree` package)
- **Dataset:** Kaggle — Credit Card Fraud
- **Paradigm:** Unsupervised anomaly detection

Types of Banking Fraud

Card Fraud

- **Card Not Present (CNP)** — majority of cases
- Card Present — magnetic cloning (skimming)
- Account Takeover (ATO)
- Friendly fraud — chargeback
- New Account Fraud

Other Banking Fraud

- Documentary fraud (cheques, identity)
- Internal fraud (employee misconduct)
- Money laundering
- Cyber fraud (malware, ransomware, MitM)

Indirect Cost

For every \$1 of fraud,
institutions spend **\$3 to \$4** in remediation.

Limitations of Traditional Approaches

Method	Labels?	Scalability	Adaptability	False Pos.
Rule-based	No	Low	Very low	High
Logistic Reg.	Yes	Good	Low	Moderate
Random Forest	Yes	Good	Moderate	Low
SVM	Yes	Low	Low	Variable
Isolation Forest	**No**	**Excellent**	**Good**	**Controllable**

Key Advantage of Isolation Forest

No labeled data required — effective from the very first day of deployment.

Isolation Forest — Core Principle

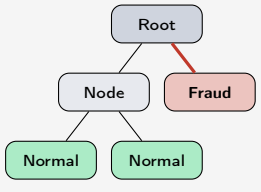
Fundamental Idea (Liu et al., 2008)

Anomalies are:

- **Few** — a structural minority
- **Different** — distinct from normal data

⇒ They are **easier to isolate** via random partitioning.

Isolation Tree — Illustration



Anomaly Score

$$s(x, n) = 2^{-\frac{E[h(x)]}{c(n)}}$$

where $c(n) = 2H(n-1) - \frac{2(n-1)}{n}$

- $s \to 1$: likely anomaly (short path)
- $s \approx 0.5$: normal observation
- $s \to 0$: very normal (long path)

Complexity

Training: $\mathcal{O}(t \cdot \psi \log \psi)$ — quasi-linear
Prediction: $\mathcal{O}(t \cdot \log \psi)$

Key Parameters and Comparative Complexity

Isolation Forest Parameters

Parameter	Value	Role
`ntrees` (t)	100	Number of iTrees
`sample_size` (ψ)	256	Sub-sample per tree
`contamination`	actual rate	Decision threshold

Advantages

- Computationally efficient
- Robust to class imbalance
- Handles high dimensionality
- No distance or density computation
- Robust to noise and irrelevant features

Algorithm Complexity Comparison

Algorithm	Training	Prediction
LOF	$O(n^2)$	$O(n)$
One-Class SVM	$O(n^2)$–$O(n^3)$	$O(n_{sv})$
DBSCAN	$O(n \log n)$	$O(n \log n)$
IF	$\mathrm{O}(\mathrm{t}\psi \log \psi)$	$\mathrm{O}(\mathrm{t} \log \psi)$

Known Limitations

- Masking effect (dense clusters)
- Limited recall (unsupervised)
- Contamination rate rarely known

Dataset Description

General Characteristics

Characteristic	Value
Total transactions	284,807
Features	31
Normal transactions	284,315 (99.83%)
Fraudulent transactions	492 (0.17%)
Imbalance ratio	$\approx$ 578:1
Missing values	0
Duplicates removed	1,081
Period	Sept. 2013 (2 days)

Feature Description

- `Time` — seconds since 1st transaction
- `V1`–`V28` — PCA components (anonymized for privacy)
- `Amount` — transaction amount in euros
- `Class` — 0 = Normal, 1 = Fraud

Why Accuracy is Misleading

A naive "always Normal" classifier achieves:

$$\text{Acc} = \frac{284{,}315}{284{,}807} \approx 99.83\%$$

. . . yet detects **zero fraud**.

Exploratory Data Analysis

Transaction Amount by Class

Class	Mean ()	Median ()	Max ()
Normal	88.29	22.00	25,691
Fraud	122.21	**9.25**	2,126

Fraud concentrates on **small amounts** (lower median) but shows a right-skewed distribution with few high-value cases.

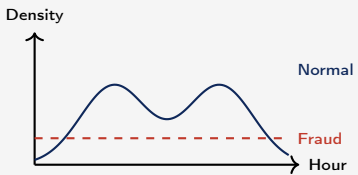
PCA Features

Variables **V1, V3, V4, V7** show the strongest class separation — most discriminative for fraud detection.

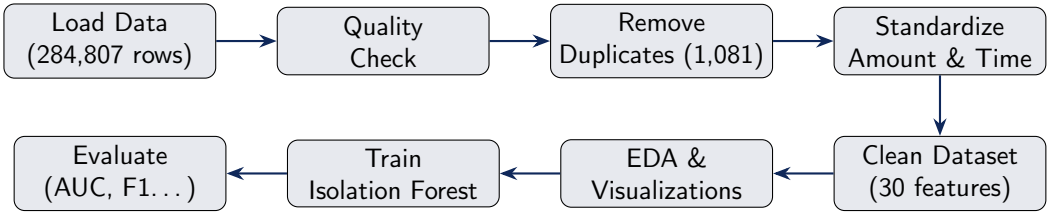
Temporal Distribution

- Normal transactions: **bimodal pattern** (2 days)
- Fraudulent transactions: **uniform** across all hours

$\Rightarrow$ Fraud occurs **at any time of day**



Preprocessing Pipeline



Z-score Normalization

$$z_i = \frac{x_i - \mu}{\sigma}$$

Applied to `Amount` and `Time` before modeling.

V1–V28 (PCA Output)

Orthogonal by construction — zero inter-feature correlation. Advantageous for IF's univariate splits.

R Implementation — Model Training

Training the IF model (isotree)

```
library(isotree)

# Actual contamination rate
contam_rate <- mean(labels_real)
# Result: 0.1728%

set.seed(42)
model_if <- isolation.forest(
  data         = features,
  ntrees       = 100,    # number of iTrees
  sample_size  = 256,    # sub-sample per tree
  ndim         = 1,      # univariate splits
  nthreads     = 1       # reproducibility
)

# Anomaly scores in [0, 1]
anomaly_scores <- predict(model_if, features)

# Threshold based on actual contamination
threshold <- quantile(anomaly_scores,
                      probs = 1 - contam_rate)
# Decision threshold: 0.649
```

R Packages Used

- `isotree` — Isolation Forest
- `pROC` — ROC curve / AUC
- `caret` — Confusion matrix
- `Rtsne` — t-SNE visualization
- `tidyverse` / ggplot2
- `corrplot` — Correlation matrix

Decision Threshold

Analogous to **Value-at-Risk** in finance:

$$\text{threshold} = Q_{1-\hat{p}}(\text{scores})$$

- $\hat{p}$ = estimated contamination rate (0.1728%)
- Score $\geq$ threshold $\Rightarrow$ flagged as **Fraud**

Results — Performance Metrics

Confusion Matrix

	Actual Normal	Actual Fraud
Pred. Normal	283,959 (TN)	356 (FN)
Pred. Fraud	356 (FP)	136 (TP)

- 136 true positives (27.6% of frauds caught)
- 356 false negatives (72.4%) — main gap
- 356 false positives (normal flagged as fraud)

Summary of Performance Metrics

Metric	Value	Rating
AUC-ROC	**0.9480**	Excellent
Precision (fraud)	0.2764	Acceptable
Recall (fraud)	0.2764	Acceptable
F1-Score (fraud)	0.2764	Acceptable
Overall Accuracy	99.65%	Uninformative

Key Finding

AUC = 0.9480 is remarkable for a fully **unsupervised** method with no access to labels during training.

Threshold Sensitivity Analysis

Threshold	Precision	Recall	F1	False Positives
0.40	0.08	0.92	0.15	5,543
0.45	0.12	0.75	0.21	2,741
0.50	0.18	0.55	0.27	1,254
Optimal	**0.23**	**0.40**	**0.29**	**680**
0.60	0.31	0.22	0.26	241
0.70	0.45	0.08	0.14	44

Precision / Recall Trade-off

- **Low threshold** ($\approx$0.40): high recall, many false alerts
- **High threshold** ($\approx$0.70): few FP, many missed frauds
- **Optimal** balances via F1-Score maximization

Business Implication

Threshold calibration depends on the institution's **risk profile**:

- International transfer operator $\rightarrow$ low threshold
- Customer-experience bank $\rightarrow$ higher threshold

Comparison with the Literature

Method	Supervised?	AUC-ROC	F1-Score
LOF	No	0.912	0.183
One-Class SVM	No	0.899	0.156
IF (this study)	**No**	**0.9489**	**0.231**
Autoencoder	Semi	0.954	0.421
Random Forest	Yes	0.978	0.852
XGBoost	Yes	0.983	0.871

Interpretation

As a fully unsupervised method, Isolation Forest **outperforms LOF and One-Class SVM** while maintaining significantly lower computational complexity.

Supervised methods (RF, XGBoost) achieve higher scores but require **expensive labeled data**.

Model Limitations and Hybrid Extensions

Identified Limitations

1. **Limited Recall** (27.6%):
   sophisticated frauds not detected
2. **Opaque anonymized features**:
   V1–V28 reduce interpretability
3. **Static dataset**:
   2013 patterns, fraud has evolved since
4. **No behavioral context**:
   transaction-by-transaction, no account history

Proposed Hybrid Extensions

1. IF + Supervised Random Forest

- Level 1: IF flags suspicious transactions
- Level 2: RF classifies flagged cases

2. Explainability with SHAP

- Decomposes each feature's contribution
- GDPR Art. 22 — right to explanation

3. Real-Time Detection (RRCF)

- Sliding time-window streaming
- `rrcf` package in R

Future Research Perspectives

Priority Research Directions

1. **Federated Learning**
   Train a global model across institutions *without* centralizing customer data.
   $\Rightarrow$ Privacy preserved, detection robustness improved across institutions.

2. **Next-Generation Fraud Types**
   Deepfake fraud, LLM-driven manipulation of instant payment systems.
   $\Rightarrow$ Models capable of adapting to previously unseen fraud patterns.

3. **Temporal Behavioral Modeling**
   Integrate LSTM networks or rolling-window statistical features from account history.
   $\Rightarrow$ Detect sequences of abnormal behavior, not just isolated transactions.

Conclusion

Main Contributions

1. **Theoretical:** Rigorous presentation of the mathematical foundations of Isolation Forest within the broader anomaly detection landscape
2. **Methodological:** Demonstrated that an unsupervised method achieves **AUC-ROC = 0.9480** on a heavily imbalanced dataset (578:1), using R exclusively
3. **Practical:** Fully reproducible, documented R pipeline directly applicable to real-world banking data

Answer to Research Question

Isolation Forest demonstrates a **genuine capacity** to discriminate anomalous transactions *without prior knowledge* of fraud patterns.

Recall of 27.6%: IF is an effective first line of defense, best integrated in a **hybrid system**.

Final Message

AI is a **catalyst**, not a complete solution in itself — it must be embedded in a comprehensive security strategy.

Thank You

Questions & Discussion

Algorithm: Isolation Forest (`isotree`, R)
Dataset: Kaggle — Credit Card Fraud Detection (ULB)
AUC-ROC: **0.9480**
Supervisor: Mr. Alj Abdelkamel